

PRASAD R

Full Stack .NET Developer

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PROFESSIONAL SUMMARY

Full Stack .NET Developer with 3+ years of professional experience building scalable, real-time software for the railway sector. Proficient in C#, ASP.NET Core, Angular, Avalonia UI, and REST API architecture. Shipped 5 production projects that serve over 50,000 daily commuters across India. Strong expertise in SQL Server and MongoDB with high-performance backend engineering. Uses ChatGPT and Claude AI to boost code quality and cut delivery time by up to 30%.

TECHNICAL SKILLS

Programming Languages: C#, SQL (T-SQL), JavaScript, TypeScript

Frontend Technologies: Angular, Avalonia UI, HTML5, CSS3, XML

Backend and Web: ASP.NET Core, ASP.NET MVC, Web API, REST API

Data Access: ADO.NET, Entity Framework, DataAdapter, DataSet, DataReader

Databases: Microsoft SQL Server 2017, MongoDB, MySQL, MS Access

Protocols and Communication: TCP/IP, RS232, RS485, Modbus

AI Tools: ChatGPT, Claude AI

Developer Tools: Visual Studio 2019, Visual Studio 2026, IIS, Git

Operating Systems: Windows, Linux

WORK EXPERIENCE

Areca Embedded Systems Private Limited

Full Stack .NET Developer | May 2023 - Present | Hyderabad, Telangana

- Engineered full stack .NET solutions for real-time commuter information and ATP systems, serving over 50,000 daily users across 12 stations.
- Architected cross-platform desktop software in C# and Avalonia UI, enabling zero-downtime deployment on both Linux and Windows production servers.
- Spearheaded Angular-based operator dashboards and monitoring portals, cutting task completion time by 40% across station teams.
- Slashed SQL Server query execution time by 35% via schema redesign and indexing strategies for high-frequency event pipelines.
- Boosted feature delivery speed by 30% by embedding ChatGPT and Claude AI into daily workflows for code reviews and technical writing.
- Streamlined REST API data flow between Central Data Controllers and output modules, sustaining 24 FPS refresh rates with under 50ms latency.
- Partnered with hardware engineers to wire software into RS485, Modbus, and serial interfaces across 5 mission-critical projects.
- Shipped 5 production-grade applications on schedule within 3 years, recording zero critical incidents after go-live across all deployments.

PROJECTS

1. Video Display Controller (VDC) - IP-based Integrated Passenger Information System

Technologies: C#, .NET Core, Avalonia UI, MongoDB, REST API, TCP/IP, Linux, Windows

- Delivered a cross-platform .NET desktop application that centrally manages IP-based Video Display Boards (VDB and AVDB) across 8+ stations simultaneously.

- Wired REST API communication between the Central Data Controller and VDC, hitting 24 FPS frame output with sub-50ms round-trip latency.
- Constructed 7 operator UI panels: Slave Communication Status, Playlist Management, Test Mode, Board Settings, and Image/Video Output views.
- Halved manual content update effort by building a multimedia scheduling engine that rotates images and videos with configurable durations.
- Sustained 99.9% uptime on Linux servers through layered error handling and auto-recovery logic tested under 24x7 load conditions.

2. Kavach Event Logger - Automatic Train Protection (ATP) System

Technologies: C#, ASP.NET, ADO.NET, Microsoft SQL Server, TCP/IP, RS232/RS485

- Logged over 10,000 ATP safety events per hour into SQL Server, meeting Indian Railways audit and compliance mandates.
- Forged a real-time data acquisition pipeline via ADO.NET and TCP/IP sockets, capturing hardware signals with under 10ms latency.
- Furnished fault analysis and event replay tools adopted by 20+ safety engineers for post-incident investigation and reporting.

3. Papis - Public Announcement and Passenger Information System (EMU/MEMU Trains)

Technologies: C#, .NET, ADO.NET, SQL Server, RS485, Serial Communication

- Launched coordinated audio-visual announcement services across 6 coaches per train unit, broadcasting over 500 notifications per day.
- Synchronized in-coach LED boards and audio units over RS485, achieving announcement delivery within 100ms of trigger events.
- Embedded automatic failover logic to guarantee uninterrupted 24x7 broadcast operation across entire EMU/MEMU fleets.

4. Integrated Passenger Information System (IPIS) - Railway Stations

Technologies: C#, ASP.NET, JavaScript, SQL Server, TCP/IP

- Deployed live arrival and departure boards refreshing 100+ terminal screens at sub-second intervals from live schedule feeds.
- Developed an operator control panel in ASP.NET and JavaScript, giving 10+ staff members the ability to override schedules and reconfigure outputs on demand.

5. Passenger Information System (PIS) and Commercial Advertisement Management

Technologies: C#, ASP.NET, SQL Server, ADO.NET

- Programmed a media rotation engine cycling 200+ advertisement slots per day across TV-based coach information units.
- Configured SQL Server schema to version 50+ media assets and automate ad scheduling with zero manual intervention.
- Built an admin portal for remote upload, preview, and scheduling, reducing content turnaround from hours to under 10 minutes.

EDUCATION

Bachelor of Technology (B.Tech) - Electronics and Communication Engineering

Avanathi Institute of Engineering and Technology, Makavarapalem, Andhra Pradesh
Jawaharlal Nehru Technological University (JNTU)

Year of Graduation: 2022 | CGPA: 7.4 / 10.0

Intermediate - Mathematics, Physics, Chemistry (MPC)

RaviTeja Junior College, Narsipatnam, Andhra Pradesh
Board of Intermediate Education, Andhra Pradesh

Year of Completion: 2018 | Percentage: 86.7%

Secondary School Certificate (SSC)

M.P.E.VEM High School, Visakhapatnam, Andhra Pradesh
Board of Secondary Education, Andhra Pradesh
Year of Completion: 2016 | GPA: 7.4 / 10.0

ADDITIONAL INFORMATION

Languages Spoken: English, Telugu

Domain Expertise: Railway Systems, ATP Software, Commuter Information Systems, Embedded Software Integration, Transportation Technology

Location: BN Reddy Nagar, Cherlapally, Secunderabad, Telangana - 500051

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.

Place: Hyderabad

Date:

Prasad R